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Raciology



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Introduction

SPECIES

- An interbreeding population which is reproductively isolated from other morphologically distinguishable groups

DEME

- A smallest population unit which is a group of genetically similar individuals bearing an intimate spatial and temporal relation

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Introduction

RACE

- An intermediate population unit existing between species population and local population
- An aggregate of local populations (Demes) of a species inhabiting a geographical Sub Division, differing taxonomically from other populations of same species
- ***It is a population of a species that differs in the frequency of some genes from other populations of the same species***

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Formation of Races

- Variations due to
  - the continuation of the process of evolution
  - adaptation to local environmental conditions
- Variations get concentrated in groups occupying a specific geography which are inter breeding
- The group will constitute specific genotypes and phenotypes and a race is born
- A race constitutes a Sub Species

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Controversies

1. No agreement on how many genetic differences it takes to make a race and hence the definition of race is ARBITRARY
2. No one race has an exclusive possession of one gene because races are OPEN
3. Race is defined as geographically associated group which make this definition REDUNDANT in today's context
4. Race is defined as taxonomically and phenotypically defined groups adapted to local environment – But doubts exist about the relationship between heredity and environment

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Controversies

5. Recent developments in anthropology about the personality and IQ – devastating conclusions – led to withholding of data – Racial concept hence is INCONCLUSIVE

6. BOYD:
   - The Criteria should be least affected by environment
   - The Criteria should not undergo mutations
   - The Criteria should have known genetic principles (Mendelism)

7. Race manifests as a cultural concept also - **Racism**

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Racial Criteria

- Morphological Criteria
  - Anthroposcopic
  - Anthropometric
- Genetic Criteria
  - Blood Groups
  - Other Genetic Markers in the populations like G6DP etc.

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Morphological Criteria

SKIN COLOR

- Melanoderms - Negroids
- Leucoderms - Caucasoids
- Xanthoderms - Mongoloids

HAIR FORM

- Leiotrichy – Straight Hair – Mongoloids
- Cymotrichy – Wavy Hair – Caucasoids
- Ulotrichy – Wooly Hair – Negroids

- Also Hair Color / Texture / Quantity / Cross Section / Whorls

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Morphological Criteria

STATURE

- Local Environment
- Nutrition
- Physical and Psychological Stress
- Ranges from VERY SHORT to VERY TALL

HEAD FORM

- Dolicocephalic (Long & Narrow)
- Mesaticephalic
- Brachycephalic (Broad & Shot)
- **Cephalic Index = (Breadth of the Head / Length of the Head) x 100**

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Morphological Criteria

FACE FORM

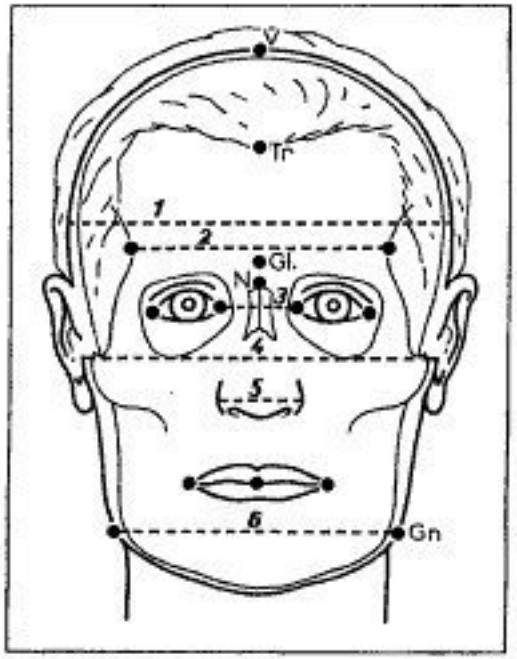
- Euryprosopic – Broad & Short
- Mesoprosopic
- Leptoprosopic – Long & Narrow
- Facial Index = (Length of the Face / Bizygomatic Breadth) x 100

NOSE FORMS

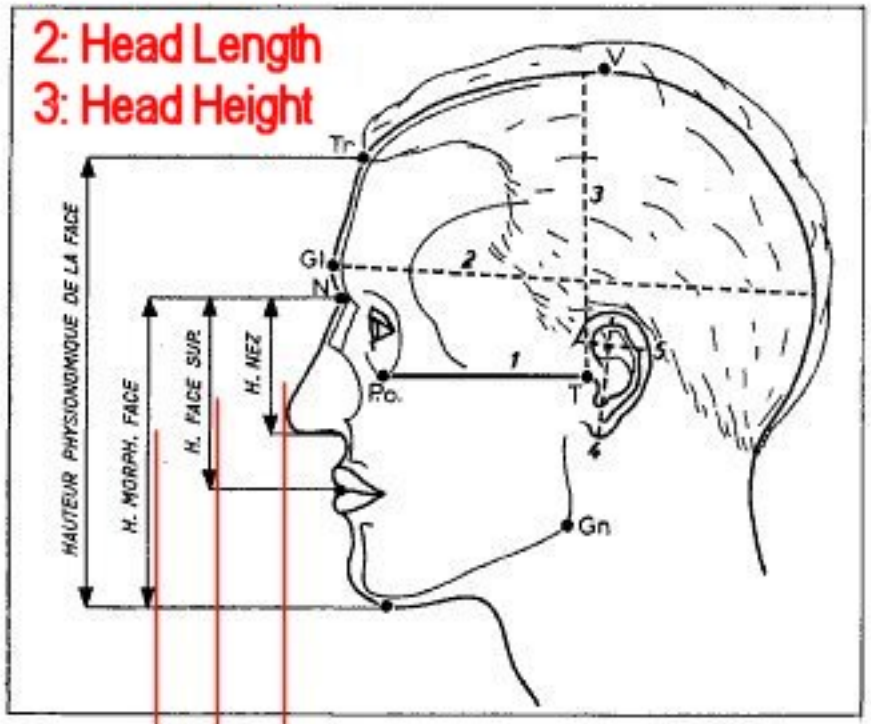
- Leptorrhine
- Mesorrhine
- Platyrrhine
- Nasal Index = (Nasal Breadth / Nasal Length) x 100
- Also Nasal Root / Nasal Bridge etc.

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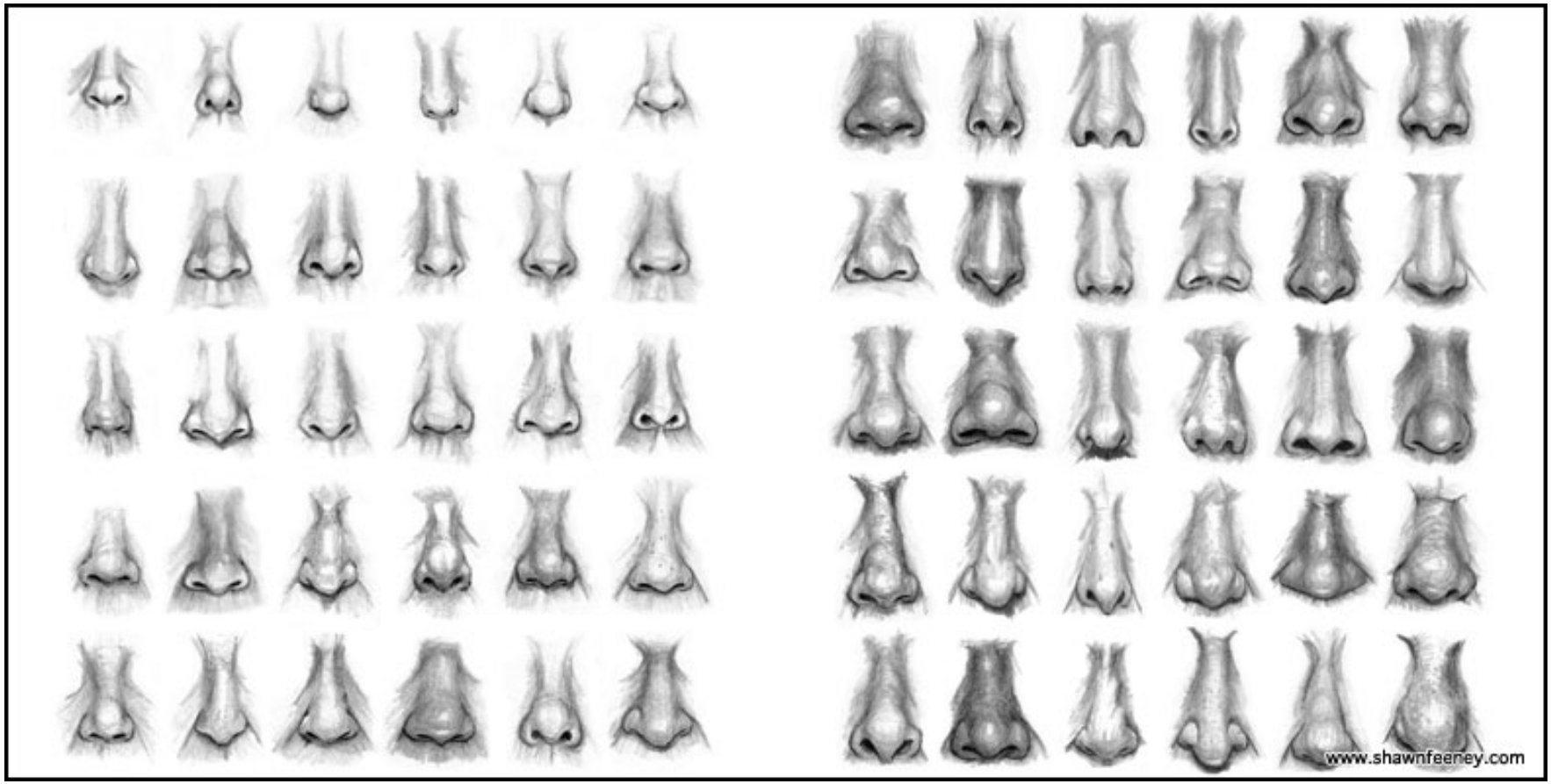
1: Head Breadth
2: Minimum Frontal
4: Bizygomatic
5: Nasal Breadth
6: Bigonial



Nasal Height
Upper Facial Height
Total Facial Height

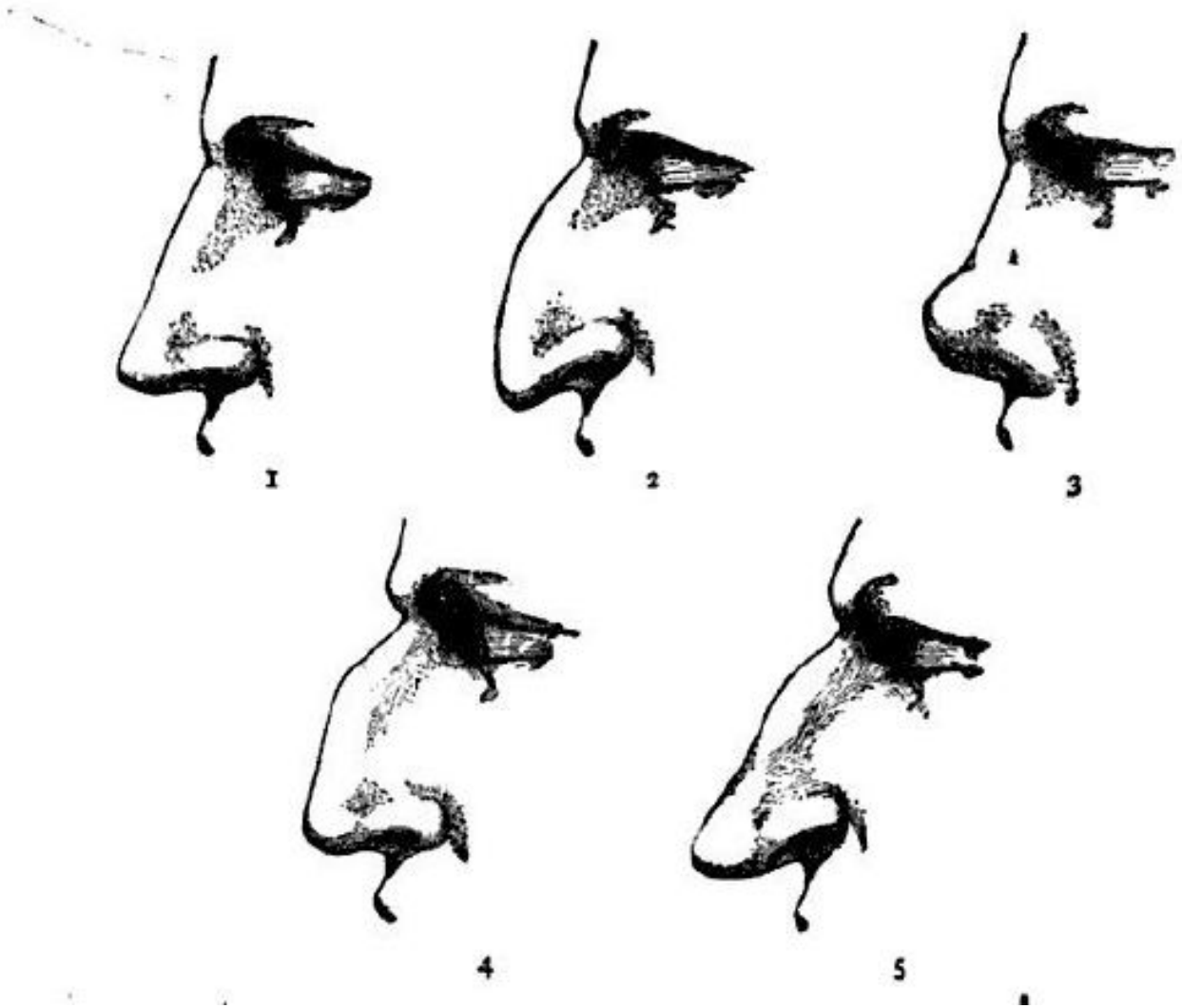
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Genetic & Serological Criteria

- Blood Groups
- Rh System
- Hemoglobin Polymorphism

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Factors in formation of race

Four Factors

- Climate
- Nutrition
- Genetic
- Culture

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Climate

Bergman's Rule

- People living in warm areas have greater body bulk than those living in cold areas

Allen's Rule

- People living in cold areas have small appendages

Glogger's Rule

- People living in hot & damp places tend to have very dark skins…

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Culture

- Socio economic factors influence genetic characters that may become race specific in the long run
- Endogamy
- Preferential Marriages
- Clan Affiliations
- Caste
- These factors influence development and hereditary retention of certain genetic characteristics

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Nutrition

- Differences in nutrition habits and status produce physical differences in the long run
- Migration leads to introduction of new foods
- Nutrition in turn is influenced by
  - Climate
  - Culture
- Too much of dependence on one staple crop leads to certain vitamin deficiencies
- These deficiencies in turn may act as selective forces within a population

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Nutrition

- All these deficiency diseases eliminate large number of populations, esp. the children and the AVERAGE BODY SIZE BECOMES SMALL
- In non lethal cases, there is a delayed growth and permanent stunting
- In cold areas, there is little or no agriculture and the chronic deficiencies leads to stunted growth

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Genetic Factors

- Environment is capable of producing genetic effects that become hereditary
  - Ex. Mutations caused by UV Radiation
- Certain peculiar mutations that are favored by natural selection might become hereditary and thus race specific
- Both heredity and environment complement each other in formation of races

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